FINAL YEAR PROJECT · 2019

Cafe Token Management System

A mobile cafeteria system for schools with parental oversight, prepaid tokens, and barcode-based transactions.

Submitted by

Asadullah

B.Sc. Information Technology · 8A · 2015-Arid-2453

Supervisor

Muhammad Ahsan

PMAS Arid Agriculture University, Rawalpindi

01 Introduction

The Cafe Token Management System is an iOS application that digitises cafeteria operations in educational institutions.

It gives students a simple way to order food, lets administrators manage menus and inventory, and allows guardians to monitor and control their children's consumption remotely.

The result: less paper, faster service, tighter financial control, and real transparency for parents.

3

User roles

Admin · Student · Guardian

iOS

Platform

Built in Swift with XCode

SQL

Backend

SQL Server 2014 via Web Services

02 Problem Statement

Traditional cafeteria operations are slow, opaque, and cash-heavy.

Paper-based workflows

Manual tokens and handwritten ledgers slow down service and create reconciliation errors.

No parental visibility

Guardians have no way to see what their children consume or how their pocket money is spent.

Limited control over diet

Parents cannot restrict unhealthy or allergen-triggering items for their child.

Slow transactions

Cash handling and queuing create bottlenecks during peak lunch hours.

03 Objectives

01 Digitalize cafeteria records of canteen sales, food items and products in a single system.

02 Replace paper-based tokens with prepaid mobile tokens redeemed via barcode.

03 Give guardians oversight and control over their children's consumption.

04 Improve transaction speed, accuracy and end-of-day reconciliation for staff.

04 Project Requirements

Admin

- Create, edit, view and delete food categories and items
- Manage student and guardian accounts
- Accept or deny account requests
- Scan barcodes at point of sale
- Generate sales and order reports

Student

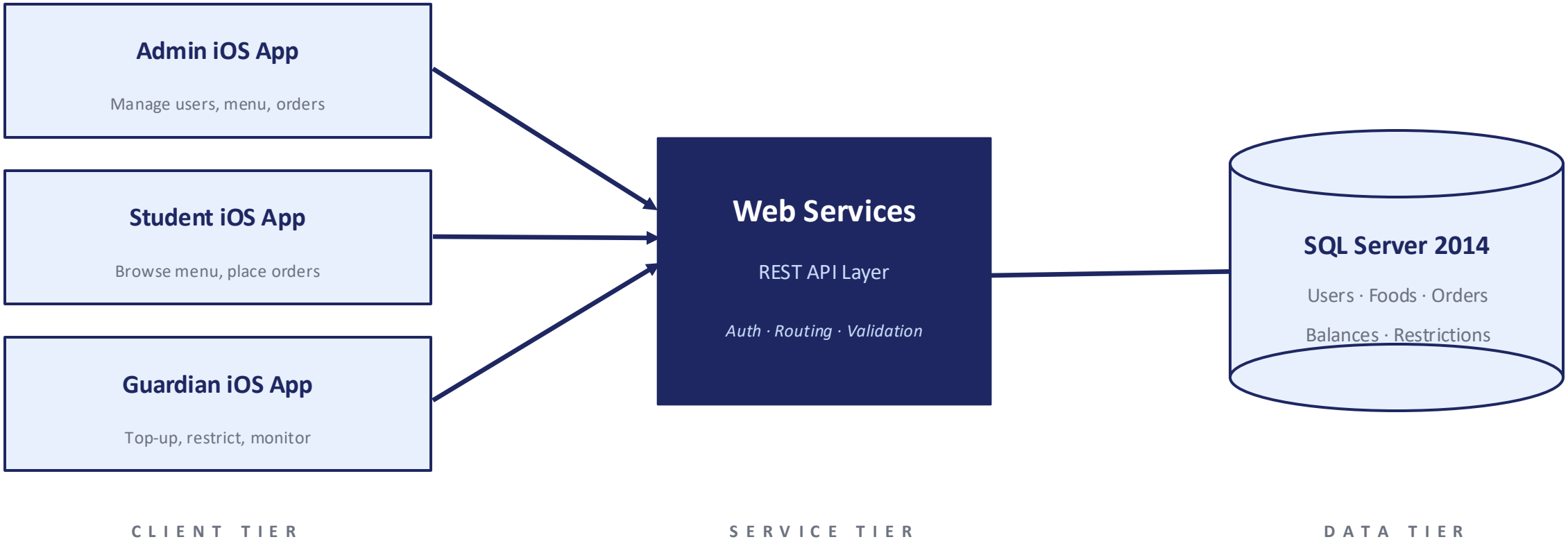
- View the cafeteria menu with prices
- Place orders and calculate totals automatically
- Receive on-screen barcode for checkout
- View order history and account balance

Guardian

- Add balance to the child's account
- Restrict specific food items for the child
- View the child's order history and spending
- Monitor the remaining token balance

05 System Architecture

Three client apps communicate with a SQL Server backend through a shared web services layer.



06 Technology Stack

LANGUAGE

Swift

iOS-native application development

IDE

XCode

Apple's official development environment

DATABASE

SQL Server 2014

Relational storage for users, orders and balances

BACKEND

Web Services

REST API layer bridging client apps and database

HARDWARE

Barcode Scanner

Fast cashier-side order verification

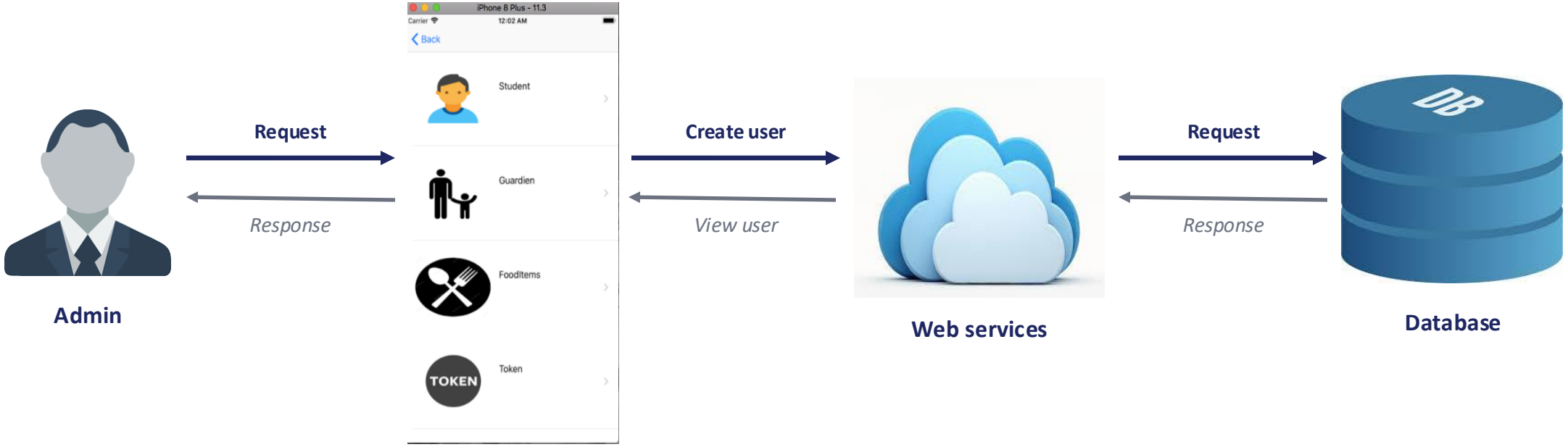
DEVICE TARGET

iPhone (iOS 11+)

Production build tested on iPhone 8 Plus

07 Conceptual Diagram — Admin

How the Admin interacts with the system to manage users and menu.

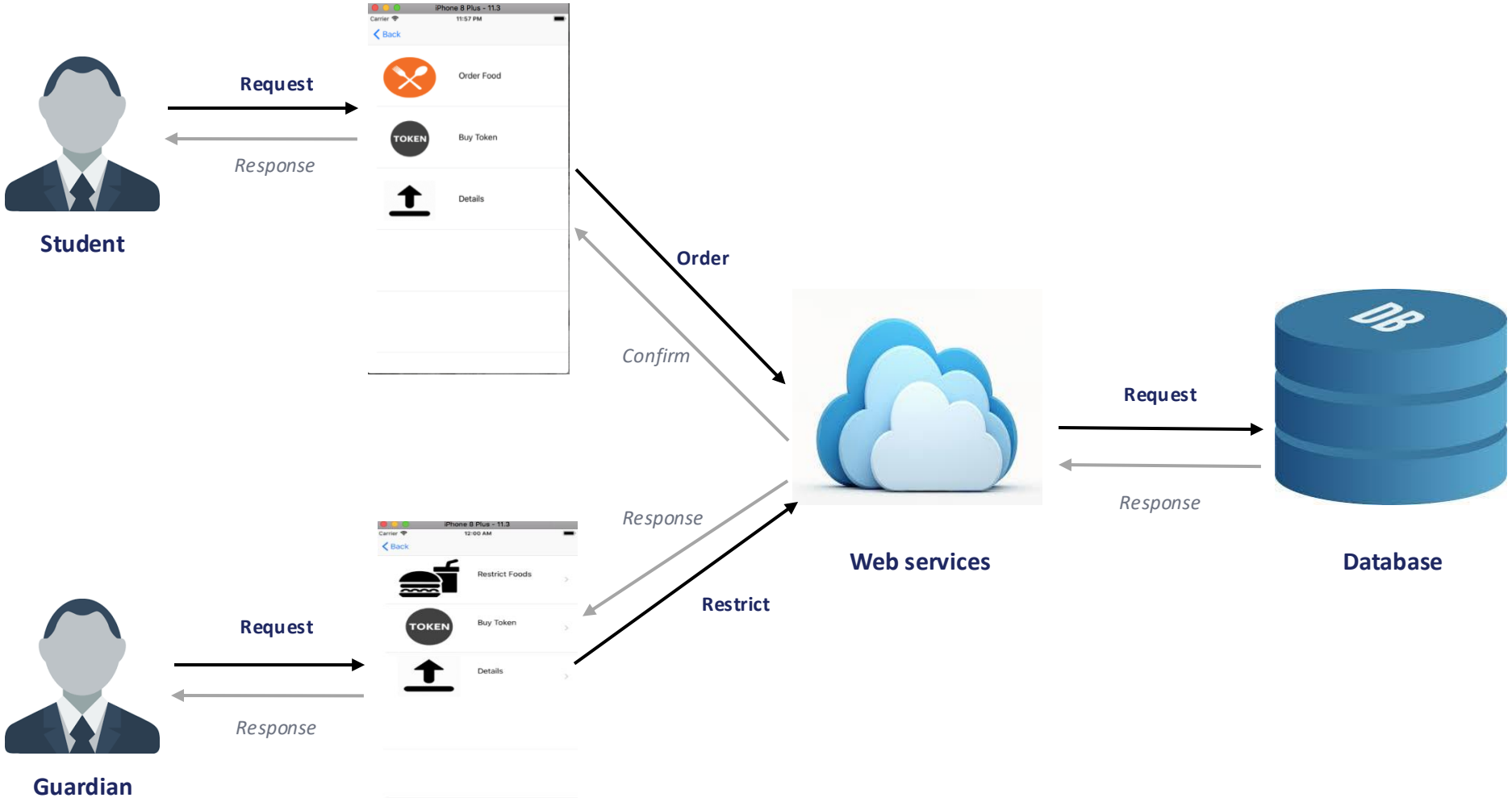


Admin Responsibilities

- Add Student
- Add Guardian
- Add Food Items
- Accept / Deny Request
- Scan Barcode

08 Conceptual Diagram — Student & Guardian

Both roles share the same service layer but access different features.



Student Actions

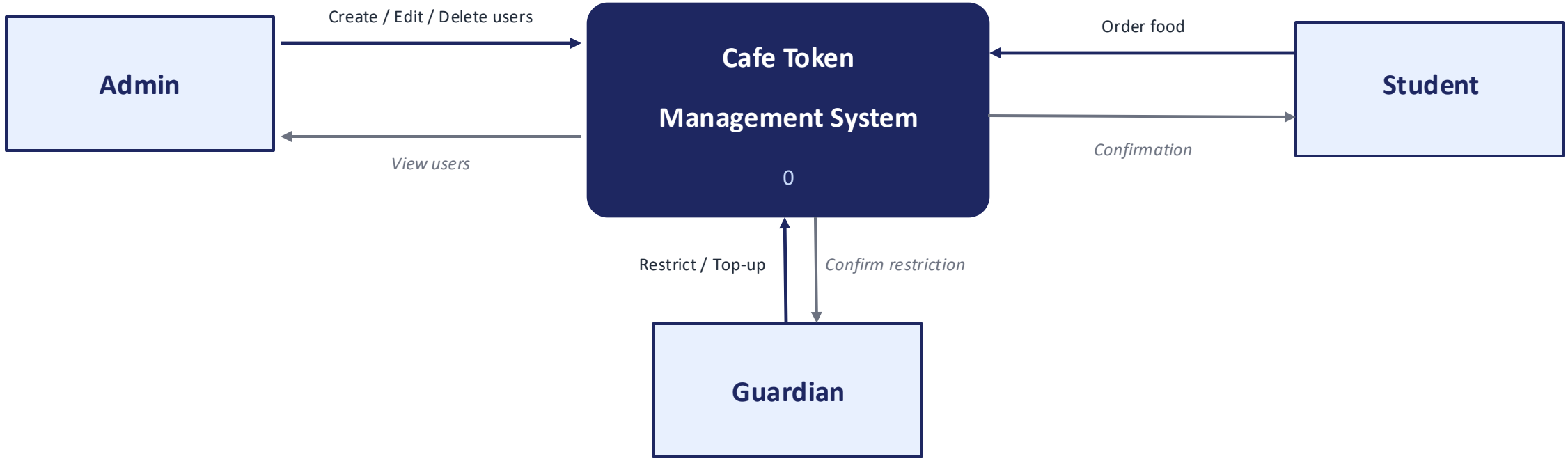
- Order food
- Add Balance
- View Order details

Guardian Actions

- Restrict Foods
- Add Balance
- View child's orders
- View balance details

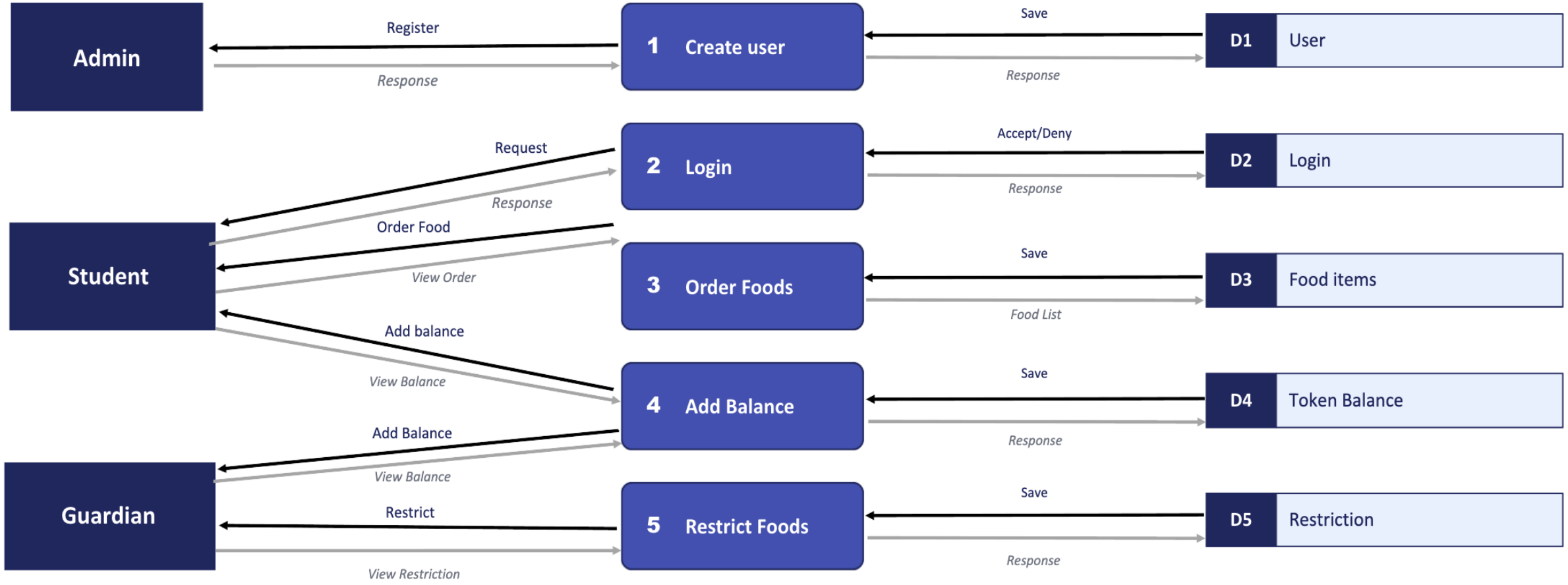
09 DFD — Context Diagram

High-level view of how each actor interacts with the system.



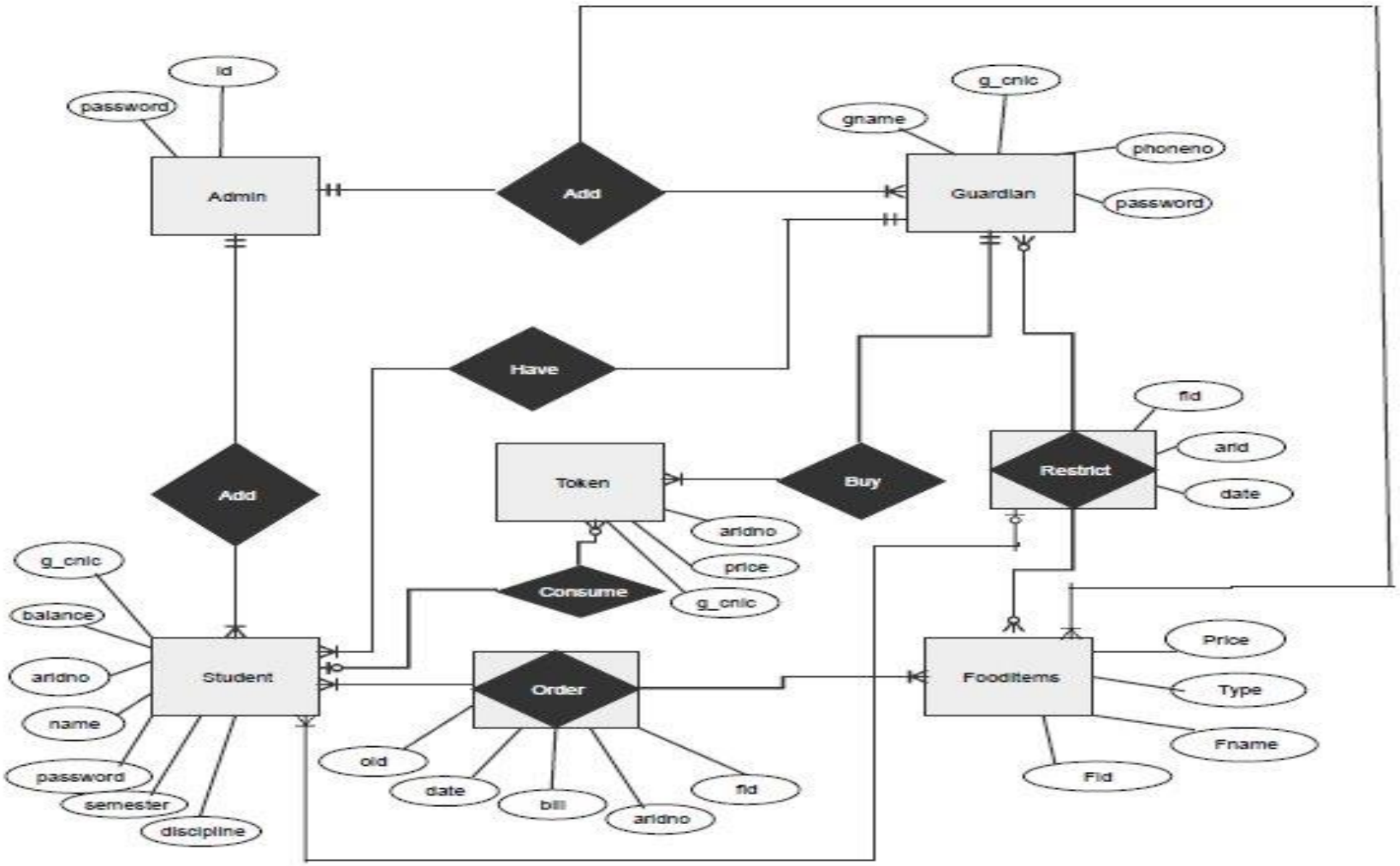
10 DFD — Level 1

Detailed data flow across the five core processes.



11 Entity Relationship Diagram

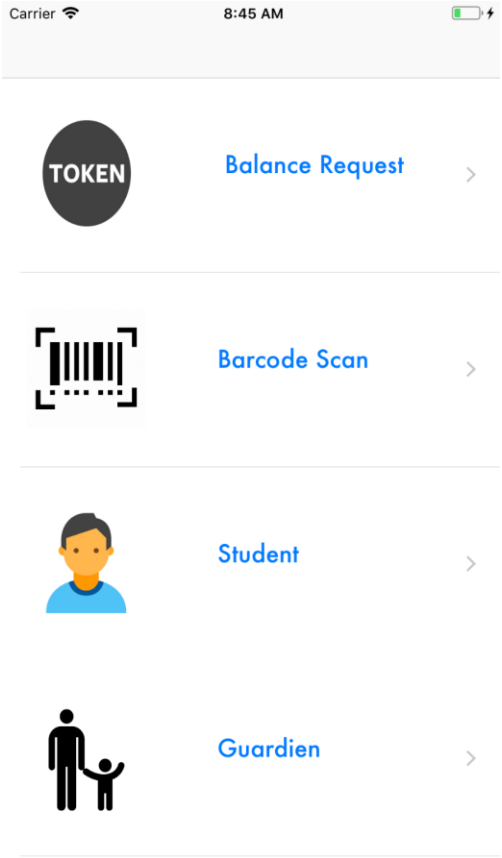
Database schema showing entities and their relationships.



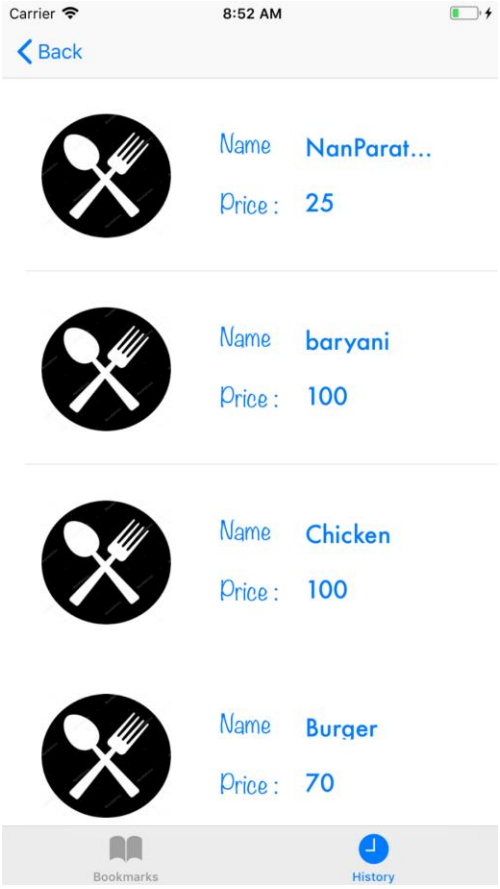
12 Admin — Screens

User management, menu administration, and barcode-based order verification.

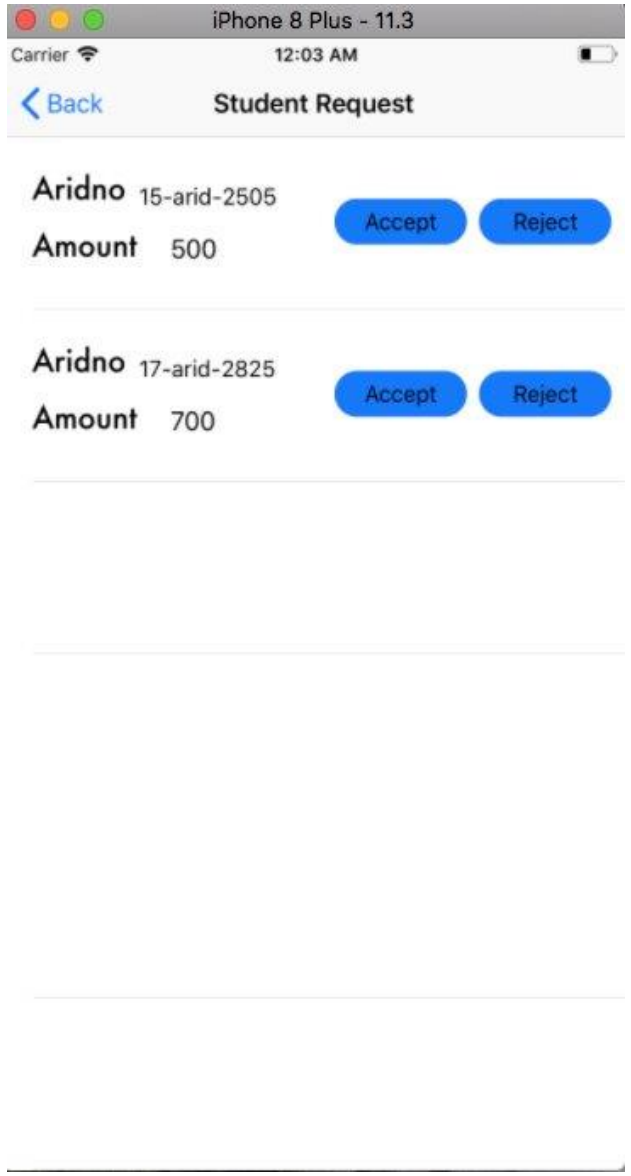
Admin Login

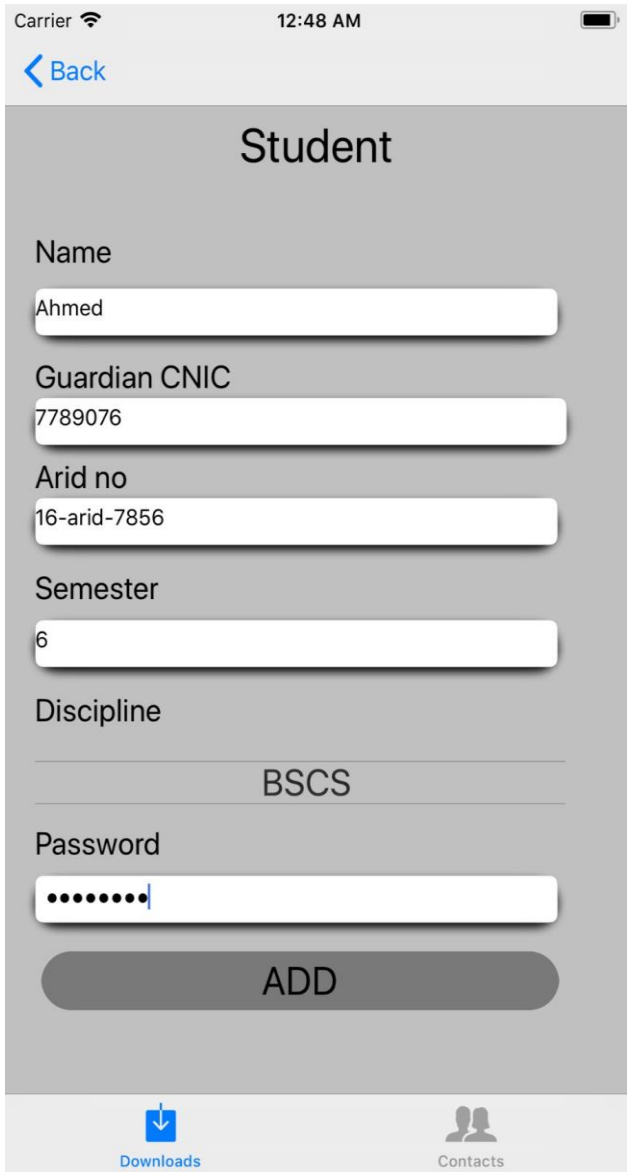


Home Dashboard



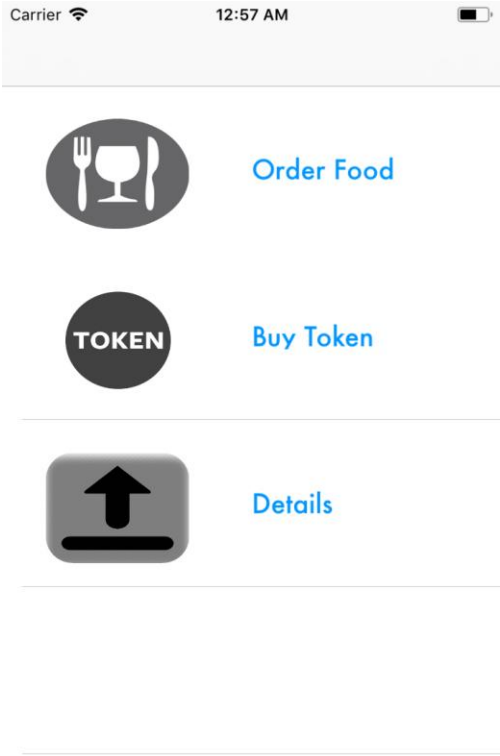
Food Menu


iPhone 8 Plus - 11.3
Carrier
12:03 AM
Back
Student Request
Aridno 15-arid-2505
Amount 500
Accept
Reject
Aridno 17-arid-2825
Amount 700
Accept
Reject

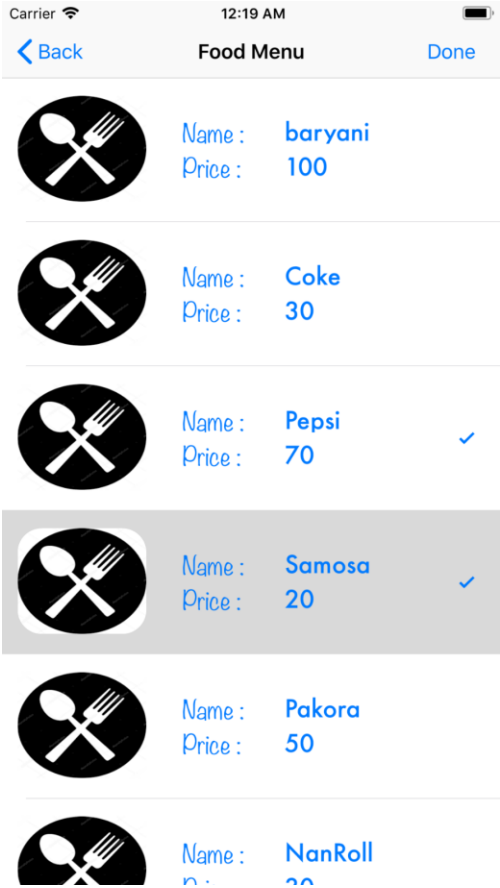

Carrier
12:48 AM
Back
Student
Name
Ahmed
Guardian CNIC
7789076
Arid no
16-arid-7856
Semester
6
Discipline
BSCS
Password
ADD
Downloads
Contacts

13 Student — Screens

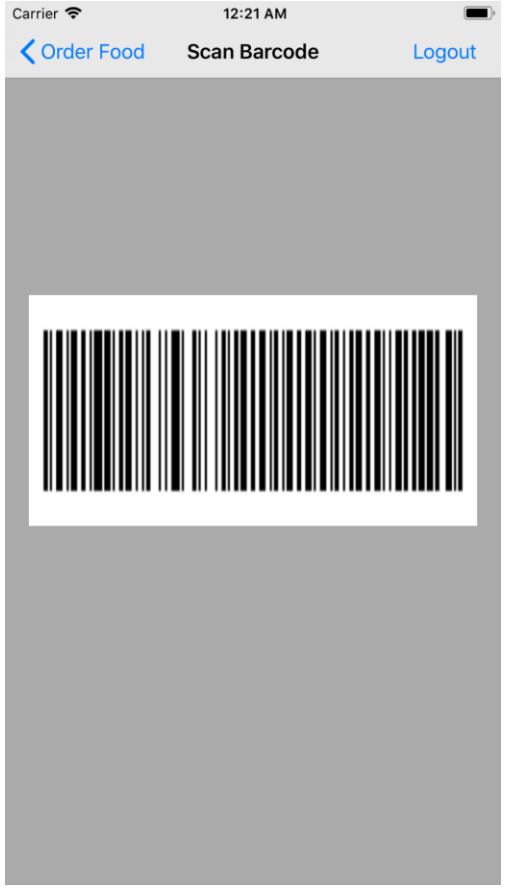
Menu browsing, order placement and barcode checkout.



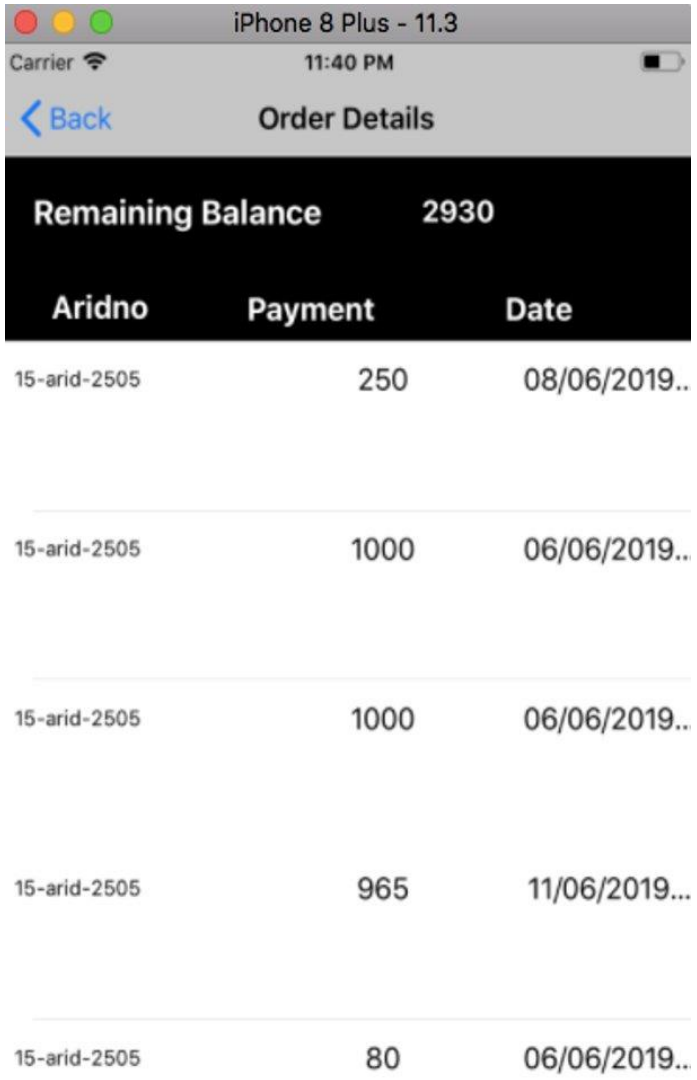
Student Home

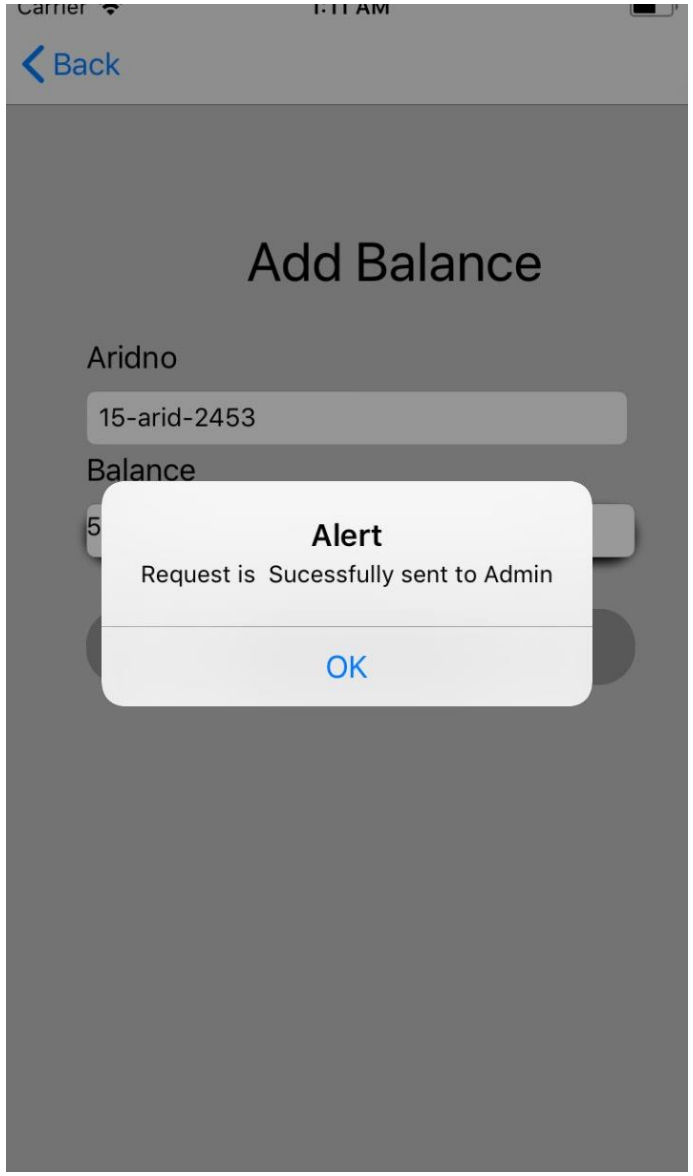


Browse Food Menu



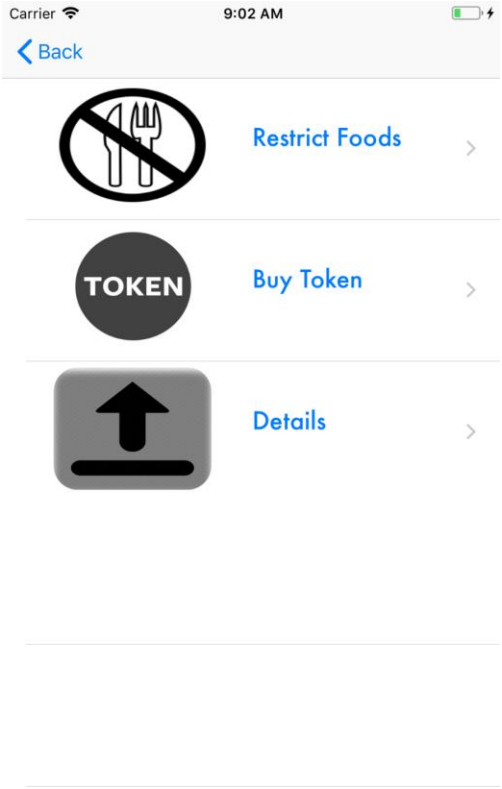
Barcode Checkout


iPhone 8 Plus - 11.3
Carrier
11:40 PM
Back
Order Details
Remaining Balance 2930
Aridno Payment Date
15-arid-2505 250 08/06/2019..
15-arid-2505 1000 06/06/2019..
15-arid-2505 1000 06/06/2019..
15-arid-2505 965 11/06/2019...
15-arid-2505 80 06/06/2019..

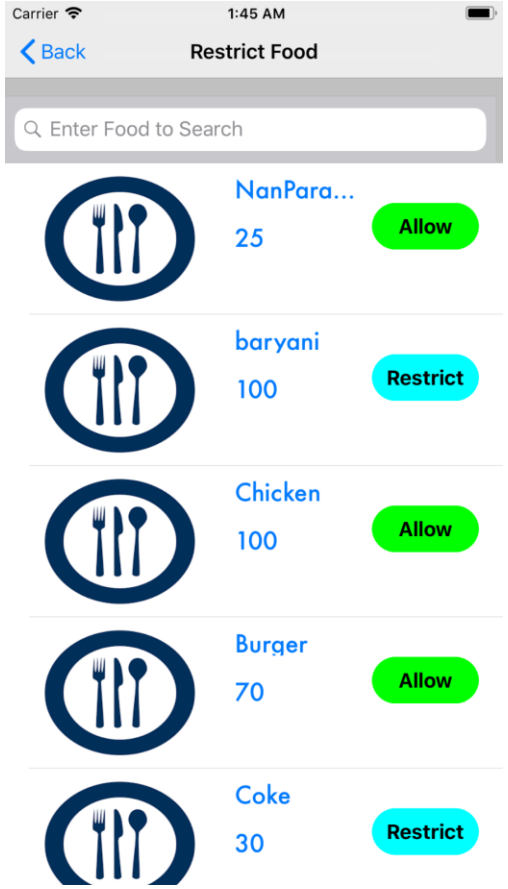

Back
Add Balance
Aridno
15-arid-2453
Balance
Alert
Request is Sucessfully sent to Admin
OK

14 Guardian — Screens

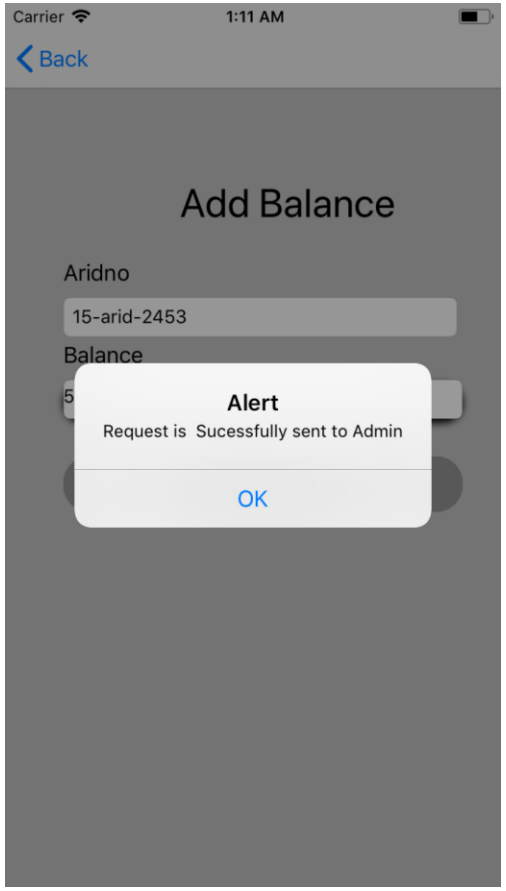
Top-up, food restriction and child activity monitoring.



Guardian Home



Restrict Foods



Add Balance

15 Key Outcomes

3

Role-based iOS clients

Admin, Student and Guardian — each with a distinct workflow.

100%

Solo implementation

Designed and built end-to-end: frontend, backend and database.

0

Paper tokens needed

Replaced fully by in-app prepaid balance and on-screen barcodes.

Live

Guardian visibility

Parents see consumption and spending in real time.

16 What I Learned

1

Full-Stack Ownership

Building frontend, backend, database and deployment end-to-end taught me the full lifecycle of a real product.

2

API Design

Designing web services for three different client apps forced me to think carefully about contracts, authentication and error handling.

3

Data Modelling

Designing the ERD from scratch sharpened my understanding of normalisation, relationships and transactional integrity.

4

User Empathy

Each of the three user roles had distinct mental models. Designing for students, parents and cafeteria staff at the same time taught me that UX is never one-size-fits-all.

THANK YOU

Questions are welcome.

Asadullah

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